

SODA-AIM_and_METHOD

Aim

The goal to characterise Antarctic subglacial landscape features through statistical and spectral analysis of bedrock topography. Historical radio-echo sounding (RES) flight lines provide bedrock elevation measurements. Contemporary Reference Elevation Model of Antarctica (REMA) is used to calculate mean ice thickness and its spatial variability. Local ice surface gradients derived from REMA are also used to infer ice flow direction.

Method (analyse_bedrock)

0. Data Ingestion & Cleaning

- **Explicit Error Handling:** The script implements `try-except` blocks for each file to ensure that a single missing or corrupted dataset does not halt the entire analysis.
- **Invalid Data Handling:** Explicitly filters invalid values for bedrock altitude and trajectory IDs.
- **Null Reporting:** The console now outputs the exact count of filtered nulls for each region to provide immediate feedback on dataset quality.
- **load_datasets() function** Implements a **dictionary-based configuration**. The user defines a list of `target_files` with labels and optional subsetting functions, making it trivial to add or remove regions. The `load_datasets()` function returns a **list of dictionaries** (`all_dfs`), each containing both the `descriptive name` and the `data`
- The `analyse_bedrock()` function loop then iterates through this list and proceeds to process each region separately to analyze the topography.

1. Data Segmentation & Pre-processing

Gap Detection: The script uses `detect_data_gaps` with a `gap_threshold=2000` (2km) to identify discontinuities.

Data Splitting: `split_into_segments` breaks each trajectory into independent chunks. This prevents the spectral analysis from "smearing" data across large physical voids.

Length Filtering: Chunks shorter than `min_segment_length=50` points are discarded to ensure data validity for the Lomb-Scargle periodogram.

Thickness Threshold: The script introduces a strict **validity check for ice thickness**. Segments are now automatically discarded if less than **20%** of the data has valid thickness measurements.

2. Spatial Domain Analysis (Per Segment)

Bedrock Detrending is applied at two scales. First, a single linear detrend is applied to the full segment to remove the large-scale slope; this is used for the spatial profile plots. Separately, within each **sliding window** (50 km, 80% overlap), a local linear detrend isolates the topographic residuals used for spectral analysis and RMS roughness calculation, as well as the morphometric statistics (local relief, RMS roughness, largest local

structures). The plots display the full-segment detrend rather than the per-window detrending.

NOTE if a segment is shorter than 50 km, the code falls back to treating the entire segment as a single window (`window_size=segment_len_m`).

REMA Surface Integration: For each segment, surface elevation is extracted from the REMA mosaic.

2.1. Flight Line Distance Calculation

The input data provides coordinates in latitude/longitude (WGS84) but lacks reliable along-track distance information. To enable spectral analysis, the script converts coordinates to Antarctic Polar Stereographic (EPSG:3031) using `pyproj` , then calculates cumulative along-track distance in metres.

2.2. Bedrock Spectral Signal Processing

refs: [Jordan_2017, Bingham_2009, VanderPlas_2018]

Frequency domain analysis for non-periodic and unevenly spaced real-world data

Welch-like averaging approach: A "rectangular" window (no tapering) is applied to each sliding segment, preserving signal energy and providing the finest frequency resolution. Although rectangular windows exhibit greater spectral leakage than tapered alternatives, this trade-off is acceptable given the power law fitting approach.

Window Size: 50 km. **Step Size:** 10 km.

Overlap: $\frac{50k-10k}{50k} = 0.8 = 80\%$ between consecutive windows.

The script includes a **fallback mechanism**: if a segment is shorter than 50 km, the code treats the entire segment as a single window.

Lomb-Scargle Periodogram: This algorithm performs least-squares spectral analysis without requiring resampling or uniform spacing. Instead, it fits sinusoids directly to the data at their original along-track positions—well suited to radar data, which is rarely evenly spaced due to variations in aircraft ground speed. The script calculates a Lomb-Scargle periodogram over a frequency grid spanning from $1/\text{window_size}$ to the Nyquist limit. The power-law fit is then restricted to a 250m–50km wavelength band to ensure consistency across regions.

then averages these to produce a robust power spectral density (PSD). A two-pass power law fit is applied to this averaged PSD.

1. **First Pass:** The script fits an initial power law to the PSD and identifies significant peaks in the residuals.
2. **Second Pass:** The two-pass fit masks peaks with a buffer of ± 5 frequency bins to isolate the background roughness spectrum, and the power law is refitted to the *cleaned* data.

Finally, the script calculates the **whitened spectrum** by dividing the full PSD by the cleaned power-law fit. This flattens the background roughness trend, making spectral peaks associated with larger bedforms more prominent.

The spectral slope (β) is saved and displayed in the final results summary. This parameter characterises the scale-dependent roughness of the terrain:

$$\text{PSD}(f) \propto f^{-\beta} \Leftrightarrow \text{PSD}(f) \propto \lambda^{\beta}$$

Hurst exponent H is a measure of surface self-affinity, is derived from the spectral slope as:

$$H = \frac{\beta - 1}{2}$$

this metric is reported in the final results to characterize surface self-affinity.

Beta β	Landscape Type	Interpretation
< 1.5	White Noise	Chaotic. Can indicate data noise, OR shattered crystalline bedrock (e.g., deep glacial troughs) [VanderPlas_2018]
~2.0	"Brownian" / Fractal	Hard Bed. Typical of crystalline shields (granite/gneiss). The ice has to "stick and slip" over these jagged teeth. Friction is high. The curve is flatter. This means there is a lot more "power/energy/amplitude" at high frequencies (small wavelengths). [Jordan_2017]
> 2.5	Smoothed / Draped	Soft Bed. Likely a sedimentary basin (mud/till). The ice can slide rapidly over this deformable sludge. This is often where you find fast-flowing Ice Streams . The curve is steep. This means the "power" drops off very fast at high frequencies. [Bingham_2009]

Note

*These classification boundaries are approximate and that β alone doesn't uniquely determine landscape class. **Relief, anisotropy, and wavelength distribution** together give a more robust classification than any single metric.*

Peak Finding: The script uses `scipy.signal.find_peaks` to identify peaks in the whitened spectrum that exceed the background trend by a factor of two or more. These peaks correspond to dominant topographic wavelengths within each flight line. Detected wavelengths are aggregated across all flight lines to assess whether a *characteristic* periodicity exists at the basin scale.

A minimum sampling floor is enforced when calculating the Nyquist limit: `max(dx_median, 15.0)`. This ensures a minimum spacing of 15m per window, preventing aliasing artifacts caused by GPS jitter or overlapping points at shorter wavelengths.

The script compiles summary statistics, calculating the **mean** and **range (min, max)** for all metrics across flight lines.

Only *confirmed* wavelengths are included in the final statistics. A wavelength is confirmed if $\lambda < L/2$, where L is the segment length. Wavelengths exceeding this threshold are classified as *candidates* and excluded, since fewer than two complete cycles fall within the observation window (making them indistinguishable from linear trends or red noise.)

The detectable range is explicitly constrained to 250m – 50km. Wavelengths longer than the observation window cannot be reliably resolved and introduce spectral leakage, causing trends to mimic periodic signals.

Note: Wavelengths $> L/2$ are excluded because they cannot be distinguished from **linear trends or red noise** within the observation window

2.3. Plotting PSD (`plot_spectra`)

This function generates a three-panel figure:

- **ax1** : Bedrock elevation profile (black line).
- **ax2** : Log-log plot of wavelength versus PSD, with the power-law fit overlaid.
- **ax3** : Whiten spectrum (residuals), used to assess whether detected wavelengths represent physical bedforms or noise.

Peak wavelengths are marked with coloured points: **blue** for the minimum detected wavelength and **red** for the maximum.

NOTE: The code limits spectral plots to the first 5 segments (`if plot_count < 5`).

3. Other Statistical analysis

3.1 Ice Thickness

The workflow calculates ice thickness by integrating local radar-derived bedrock data with the **Reference Elevation Model of Antarctica (REMA)** surface mosaic.

Surface Elevation Extraction

Surface elevations are extracted from the REMA DEM at specific flight track coordinates using **Bilinear Interpolation**.

```
from scipy.ndimage import map_coordinates
# Interpolate:
# order=1 is Bilinear (linear in x, linear in y)
# mode='nearest' handles edges gracefully
elevations = map_coordinates(data, [rows, cols], order=1, mode='nearest')
```

- **Coordinate Mapping:** Projected world coordinates (EPSG:3031) are mapped to image pixel indices using an inverse affine transform.
- **Interpolation:** A first-order spline interpolation (`order=1`) calculates the surface height at the exact point of the radar measurement, ensuring high spatial accuracy that matches JR's Fortran-based method.

Thickness Derivation

Ice thickness (H) is determined by the difference between the interpolated surface elevation (z_s) and the radar-measured bedrock altitude (z_b):

$$H = z_s - z_b$$

Data Validation

To ensure physical consistency, the script applies two primary filters:

- **NoData Handling:** REMA "nodata" values are converted to NaNs to prevent interpolation artifacts.
- **Physical Constraint:** Any calculated thickness values < 0 (where the measured bed appears higher than the surface) are set to NaN to maintain dataset integrity.

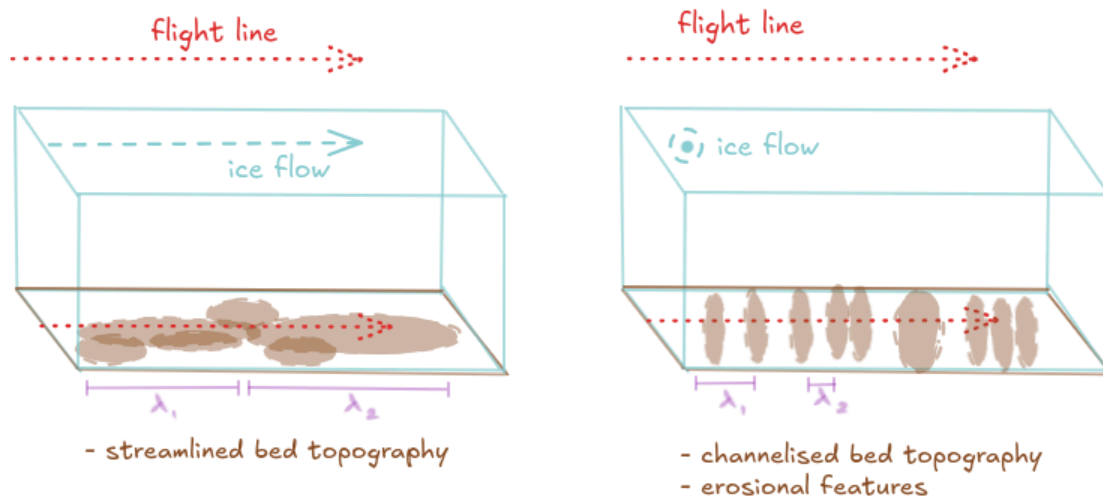
3.2 Determining "True" Flow Direction For Each Flight Line (Anisotropy)

ref [McCormack_2019]

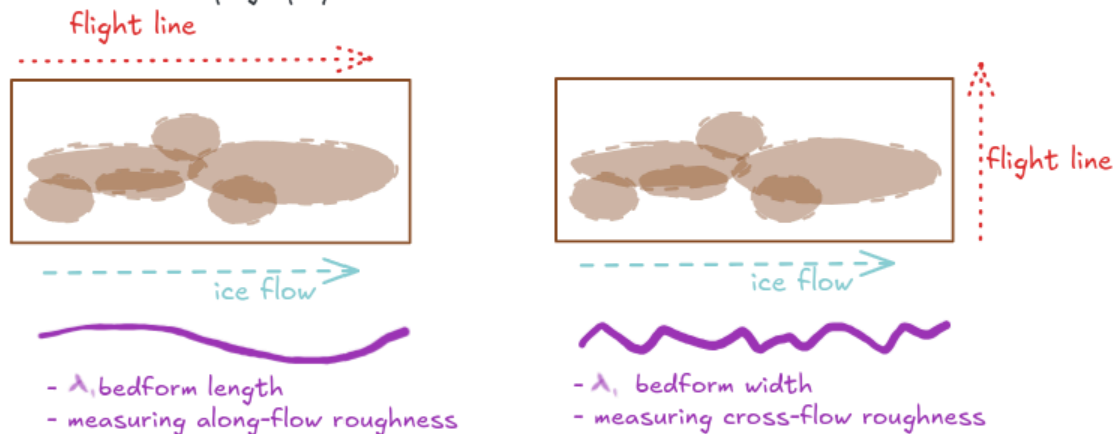
The physical interpretation of detected wavelengths depends on local ice flow direction, because bedrock roughness is anisotropic—it varies with measurement orientation. As described in Section 2.2, the spectral slope β characterises the scale-dependent roughness of the terrain, and its value varies systematically with flight line orientation relative to ice flow:

- **Parallel to ice flow:** Flight lines sample along-flow roughness. Higher β values indicate reduced high-frequency power, consistent with streamlined, glacially smoothed terrain.
- **Perpendicular to ice flow:** Flight lines sample cross-flow roughness. Lower β values indicate greater high-frequency power, reflecting rougher, less modified terrain.

(a) different bed topography



(b) identical bed topography



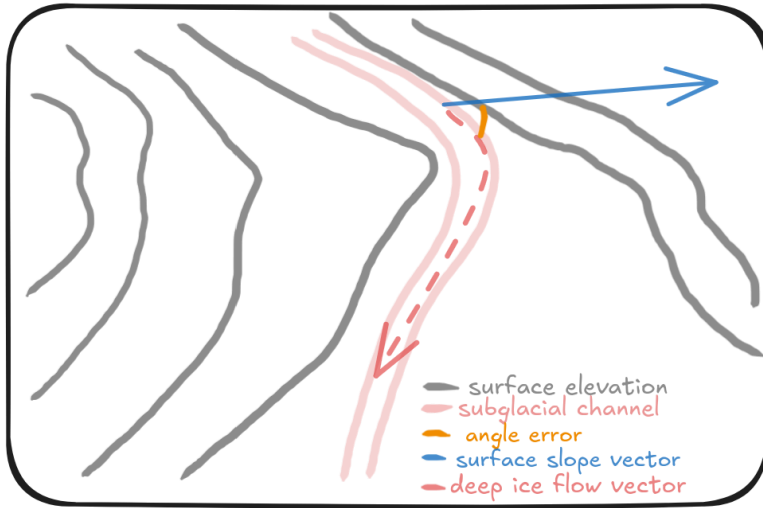
 Schematic illustrating the anisotropic interpretation of detected wavelengths.

- (a) Different beds, same flight line orientation: detected wavelengths (λ) differ due to genuine geological variation.
- (b) Same bed, different flight line orientation: detected wavelengths differ because orthogonal transects sample different dimensions of the same features.

A detected wavelength may therefore represent either **bedform length** — when the flight line is parallel to ice flow — or **bedform spacing** — when the flight line is perpendicular to ice flow.

Local Surface Gradient

In deep, fast-flowing channels, the direction the **surface is tilting** (slope) is often **NOT** the same direction the **bottom ice is moving** (flow).



McCormack et al. (2019) addressed this mismatch by smoothing the surface DEM with a spatially varying triangular filter of width $8\text{--}10 \times H$ (ice thickness). This filtering removes local surface undulations that are decoupled from the bed, leaving only the broad regional slope that actually drives deep ice flow. Crucially, the smoothing is used **only** to derive flow orientation — it does not modify the bedrock elevations used in the spectral power-law fit. The smoothed slopes showed strong agreement with observed satellite velocity fields (MEaSURES v2).

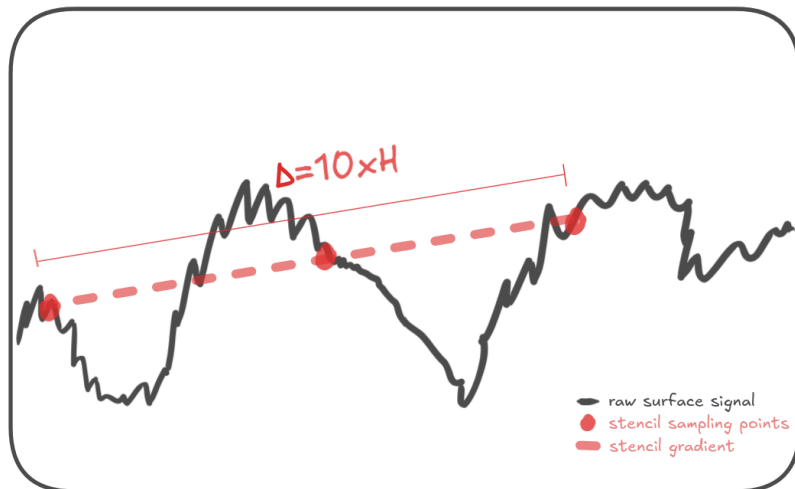
Here, the regional **ice flow vector** is estimated from the REMA mosaic (100 m resolution) using the surface elevation gradient:

$$v_{flow} \propto - \left(\frac{dS}{dx}, \frac{dS}{dy} \right)$$

where S is ice surface elevation. A **central finite difference stencil** captures the large-scale flow direction by sampling the elevation at four neighbouring points:

$$\begin{aligned} \frac{\partial S}{\partial x} &\approx \frac{z_{i+1,j} - z_{i-1,j}}{2\Delta x}, \\ \frac{\partial S}{\partial y} &\approx \frac{z_{i,j+1} - z_{i,j-1}}{2\Delta y}, \end{aligned}$$

Where $\Delta x = \Delta y = 5 \times H$.



This stencil acts as a **low-pass filter**, ignoring roughness features smaller than the stencil width. It achieves a similar goal to McCormack's triangular smoothing — suppressing local noise to find regional flow — but is faster to calculate. Gradients are sampled at flight line coordinates, allowing the **incidence angle** between each flight path and the local ice flow direction to be calculated.

Note: Flat regions are handled by imposing a unit magnitude when the gradient is zero-valued.

Flight Line Classification

The script categorises flight lines by orientation relative to flow using the dot product to calculate angles between the flight line and flow direction, ignoring upstream/downstream sense. Orientation thresholds are:

Parallel $< 30^\circ$, **Oblique** $30\text{--}60^\circ$, and **Perpendicular** $> 60^\circ$.

The analysis below also includes plots that provide spatial context for each track as a REMA hillshade background together with the analysed flightline for this region as a track colored by segment orientation class. The blue flow vectors showing ice flow direction at the surface, each segment has the average flow angle displayed as a label.

References:

@Article{Jordan_2017,

author = {T M Jordan and M A Cooper and D M Schroeder and C N Williams and J D Paden and M J Siegert and J L Bamber},

title = {{Self-affine subglacial roughness: consequences for radar scattering and basal water discrimination in northern Greenland}},

journal = {The Cryosphere},

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doi = {10.5194/tc-11-1247-2017}

}

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author = {R G Bingham and M J Siegert},
title = {{Quantifying subglacial bed roughness in Antarctica: implications for ice-sheet dynamics and history}},
journal = {Quaternary Science Reviews},
volume = {28},
number = {3},
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year = {2009},
note = {Special Theme: Modern Analogues in Quaternary Palaeoglaciological Reconstruction (pp. 181-260)},
issn = {0277-3791},
doi = {<https://doi.org/10.1016/j.quascirev.2008.10.014>},
}

@article{VanderPlas_2018,
author = {J T VanderPlas},
title = {{Understanding the Lomb–Scargle Periodogram}},
year = {2018},
month = {may},
publisher = {The American Astronomical Society},
volume = {236},
number = {1},
pages = {16},
journal = {The Astrophysical Journal Supplement Series},
doi = {10.3847/1538-4365/aab766},
}

@article{McCormack_2019,
author = {F S McCormack and J L Roberts and L M Jong and D A Young and L H Beem},
title = {{A note on digital elevation model smoothing and driving stresses}},
journal = {Polar Research},
volume = {38},
pages = {3498},
year = {2019},
doi = {10.33265/polar.v38.3498},
}